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Project Title:	Molecular Genetics of Root Thigmoresponsiveness in Arabidopsis thaliana		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Zip Code:	53706-1580	**Congressional District:**	2
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Directional cues within the environment (gravity, touch, light etc) guide plant organs' growth by differential regulation of cell expansion. These tropic movements are coupled to endogenous oscillating movements (nutations) whose origin remains poorly understood. Together, these movements allow plant organs to position themselves for optimal performance. To better understand the mechanisms that govern integrated growth responses to multiple stimuli, we have sought and characterized Arabidopsis thaliana mutations that affect root waving and skewing on tilted hard agar surfaces. Such movements result from a combination of gravity, touch, hydrotropic and other surface-derived stimuli, as well as circumnutations. One of these mutations, named wvd2-1, was shown to alter spiral root growth on agar surfaces and anisotropic cell expansion. The WVD2 protein appears to modulate the arrangement of cortical microtubules in expanding cells. This proposal is aimed at understanding how WVD2 and other related proteins modulate anisotropic cell expansion and root growth behavior. This will be accomplished by identifying and characterizing proteins that

Task Description:	regulate or modulate WVD2 function, and by characterizing seven other WVD2-related proteins in Arabidopsis. Plants constitute an essential component of Advanced Life Support Systems for long-term space-exploration missions. However, during space flight, their growth will be guided by cues other than gravity, including nutations. Therefore, we need to better understand some of the mechanisms that will contribute to guiding organ growth under these conditions, a primary objective of this proposal. Also, there is limited space available for plant growth in spacecrafts. WVD2 may provide a good tool to design reversible engineering strategies that would confer a shorter stature to a broad range of plants, yet would not alter seed production or plant's ability to perceive gravity at their final destination. Hence, a thorough characterization of WVD2-dependent pathways is of direct relevance to the Space Biology Program. The objective of this proposal is to characterize the molecular mechanisms that allow WVD2 and related WDL proteins to control root growth behavior in response to mechanical stimuli, anisotropic cell expansion and plant morphogenesis. Our long-term goal is to use the information that will derive from this project to engineer shorter and stockier plants than wild type, which can still produce normal levels of seeds, but are better suited for growth in the limited confines of bioregenerative life support systems during long-term space exploration missions, or that better resist to lodging by wind and rain on Earth or other planets. The WVD2 and WDL genes appear to function not only as key regulators of root growth behavior in response to mechanical stimuli, but also as major determinants in the control of anisotropic cell expansion in all organs of the plant. Therefore, a better understanding of these genes is likely to provide important new insights into how mechanical information in the environment controls cell expansion, and how this affects overall plant morphogenesis, a process of fundamental importance for agricultural production and plant adaptation to its environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Root-growth behavior conditions the ability of plants to populate and grow in very heterogenous and diverse environments. Hence, a better understanding of the molecular mechanisms that govern this process is likely to provide the tools needed to engineer plants that are better able to colonize extreme environments on Earth, and on other planets. At the same time, controlled anisotropic cell expansion is an essential component of plant morphogenesis and behavior in response to directional environmental stimuli. Our studies have allowed the identification of an important new dominant regulator of this process. Because over-expression of WVD2 results in shorter plants with thicker organs, it may be possible to use this engineering strategy to generate crop plants that are more resistant to lodging by wind and rain, as well as plants that are better suited for utilization in spacecrafts, where limited amounts of space are available for plant growth.
Task Progress:	The semi-dominant wvd2-1 mutation of Arabidopsis thaliana was previously isolated for its effect on root growth behavior on tilted hard-agar surfaces. Mutant roots do not wave on such surfaces, and skew in the opposite direction of gravity relative to wild type. Importantly, wvd2-1 mutant seedlings and plants display shorter and thicker organs while still producing wild type numbers of normal-looking seeds, suggesting a potential use of this system in the design of smaller plants that are better suited for growth in bioregenerative life support systems for long-term space exploration missions. Wvd2-1 was shown to derive from over-expression of a small wild type protein that appears to bind cortical microtubules in expanding plant cells. Furthermore, in vitro produced GST-tagged WVD2 appears to promote the bundling of microtubules in vitro, suggesting a mechanism by which it modulates anisotropic cell expansion and growth behavior in plants. WVD2 is a novel protein that contains a conserved domain found in seven other proteins of Arabidopsis thaliana (the WVD2-Like proteins, or WDLs). This year, we have continued our molecular analysis of WVD2 function in plants. Using transgenic WVD2-GFP fusions, we were able to image WVD2 association with the cortical microtubules in vivo. We also showed that wvd2-1 mutant seedlings display enhanced sensitivity to daily touch stimulation, a phenotype that may be attributed to microtubules with enhanced stability. Similarly, wvd2-1 seedlings were shown to display enhanced sensitivity to high light treatments. The latter observation, along with other phenotypes displayed by wvd2-1, may suggest a role for WVD2 in brassinosteroid signaling. Other WDL proteins may also control anisotropic cell expansion and cortical microtubule dynamics in Arabidopsis. Indeed, over-expression of most of these WDL proteins results in Wvd2-1-like phenotypes. However, plants homozygous for T-DNA insertion alleles of these WDL genes (knockdown or knockout alleles) display different phenotypes when grown in the presence of drugs that destabilize the microtubules, suggesting that these proteins play only partially redundant roles in the control of anisotropic cell expansion and root growth behavior in Arabidopsis. Work is underway to determine if WVD2 functions in brassinosteroid transduction, and to assign specific roles to individual WDL proteins in plant growth and development. These studies will have important implications for the design and implementation of strategies aimed at generating plants that are better suited for utilization in bioregenerative life support systems for long-term space exploration missions.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Yuen, C.Y.L., Sedbrook, J.C., Perrin, R.M., Carroll, K.L., and Masson, P.H. "Loss-of-Function Mutations of ROOT HAIR DEFECTIVE3 Suppress Root Waving, Skewing and Epidermal Cell File Rotation in Arabidopsis thaliana (L.) Heinh. " Plant Physiol, in press , Jun-2005